

JEE MAINS & IIT JEE ADVANCED

MATHEMATICS

TRIGONOMETRIC EQUATIONS

DIBY (DO IT BY YOURSELF)

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DIBY-01

Solve : $\sqrt{3}\sec 2\theta = 2$.

Ans. $\theta = n\pi \pm \frac{\pi}{12}, n \in \mathbb{I}$

DIBY-02

Find general solution of equation $\cot x = \sqrt{3}$

(A) $x = n\pi + \frac{\pi}{3}$

(B) $x = n\pi + \frac{\pi}{6}$

(C) $x = n\pi + \frac{2\pi}{3}$

(D) $x = \frac{9\pi}{6} \frac{5\pi}{6}$

Ans. B

DIBY-03

The number of values of x in the interval $[0, 5\pi]$ satisfying the equation $3\sin^2 x - 7\sin x + 2 = 0$ is:

(A) 0

(B) 2

(C) 6

(D) 8

Ans. C

DIBY-04

General solution of the equation $(2\sin\theta - \sqrt{3})^2 + (\tan\theta + \sqrt{3})^2 = 0$

Ans. $(6n + 2)\frac{\pi}{3}, n \in \mathbb{I}$

DIBY-05

Find the principal solution of $\sin 2x = -\frac{1}{2}$

(A) $\frac{7\pi}{6}, \frac{11\pi}{6}$

(B) $\frac{7\pi}{12}, \frac{11\pi}{12}, \frac{15\pi}{12}, \frac{19\pi}{12}$

(C) $\frac{7\pi}{12}, \frac{11\pi}{12}, \frac{19\pi}{12}, \frac{23\pi}{12}$

(D) None of these

Ans. C

DIBY-06

Find the number of solution $\tan 3x = 1$ in the interval $x \in \left[-\frac{2\pi}{3}, \frac{5\pi}{6}\right]$.

Ans. 05

DIBY-07

Find the number of solution of the equation $16^{\sin^2 x} + 16^{\cos^2 x} = 10$, where $0 \leq x \leq 2\pi$.

Ans: 08

DIBY-08

Example 9: Number of values of ' x ' in $(-2\pi, 2\pi)$ satisfying the equation

$2^{\sin^2 x} + 4 \cdot 2^{\cos^2 x} = 6$ is:

- (A) 8 (B) 6 (C) 4 (D) 2

Ans: C

DIBY-09

The most general solution of $\tan\theta = -1$ and $\cos\theta = \frac{1}{\sqrt{2}}$ is

- (A) $n\pi + \frac{7\pi}{4}, n \in \mathbb{I}$ (B) $n\pi + (-1)^n \frac{7\pi}{4}, n \in \mathbb{I}$
 (C) $2n\pi + \frac{7\pi}{4}, n \in \mathbb{I}$ (D) None of these

Ans: C

DIBY-10

If $4\sin^2\theta + 2(\sqrt{3} + 1)\cos\theta = 4 + \sqrt{3}$, [where $n \in \mathbb{I}$] then the general value of θ is

- (A) $2n\pi \pm \frac{\pi}{3}$ (B) $2n\pi + \frac{\pi}{4}$ (C) $n\pi \pm \frac{\pi}{3}$ (D) $n\pi - \frac{\pi}{3}$

Ans: A

DIBY-11

Solve for x : $\sin x \left(\cos \frac{x}{4} - 2\sin x \right) + \left(1 + \sin \frac{x}{4} - 2\cos x \right) \cos x = 0$

Ans: $2\pi(4n + 1)$

DIBY-12

The equation $\sqrt{3}\sin x + \cos x = 4$ has

- | | |
|-------------------------------|-------------------|
| (A) Only one solution | (B) Two solutions |
| (C) Infinitely many solutions | (D) No solution |

Ans: D

DIBY-13

The number of solutions of $\sin\theta + 2\sin 2\theta + 3\sin 3\theta + 4\sin 4\theta = 10$ in $(0, \pi)$ is

Ans:

DIBY-14

The general solution of $8\tan^2 \frac{x}{2} = 1 + \sec x$ is

- | | |
|--|-----------------------------------|
| (A) $x = 2n\pi \pm \cos^{-1}\left(\frac{-1}{3}\right)$ | (B) $x = 2n\pi \pm \frac{\pi}{6}$ |
| (C) $x = 2n\pi \pm \cos^{-1}\left(\frac{1}{3}\right)$ | (D) None of these |

Ans: C

DIBY-15

Number of solutions of $\sum_{r=1}^5 \cos rx = 5$ in the interval $[0, 4\pi]$ is:

- (A) 0 (B) 2 (C) 3 (D) 7

Ans: C

DIBY-16

$$\frac{\sin 3\theta}{2\cos 2\theta + 1} = \frac{1}{2} \text{ if}$$

- (A) $\theta = n\pi + \frac{\pi}{6}, n \in I$ (B) $\theta = 2n\pi - \frac{\pi}{6}, n \in I$
 (C) $\theta = n\pi + (-1)^n \frac{\pi}{6}, n \in I$ (D) $\theta = n\pi - \frac{\pi}{6}, n \in I$

Ans: C

DIBY-17

If $0 \leq x < \frac{\pi}{2}$, then the number of values of x for which $\sin x - \sin 2x + \sin 3x = 0$, is

- (A) 2 (B) 1 (C) 3 (D) 4

Ans: A

DIBY-18

If the sum of solutions of the system of equations $2\sin^2\theta - \cos 2\theta = 0$ and $2\cos^2\theta + 3\sin\theta = 0$ in the interval $[0, 2\pi]$ is $k\pi$, then k is equal to .

Ans: 03

DIBY-19

Solve : $\sin(\cos x) = \cos(\sin x)$

Ans: No Solution

DIBY-20

Solve : $\operatorname{cosec} x - \operatorname{cosec} 2x = \operatorname{cosec} 4x$

Ans: $x = (2n + 1)\frac{\pi}{7}$ & $n \neq 7m + 3, n \in \mathbb{I} \text{ \& } m \in \mathbb{I}$

DIBY-21

The general value of θ satisfying the equation $2\sin^2\theta - 3\sin\theta - 2 = 0$, is

(JEE – 1995, 2M)

(A) $n\pi + (-1)^n \frac{\pi}{6}$
 (C) $n\pi + (-1)^n \frac{5\pi}{6}$

(B) $n\pi + (-1)^n \frac{\pi}{2}$
 (D) $n\pi + (-1)^n \frac{7\pi}{6}$

Ans: D

DIBY-22

The general solution of $\sin x - 3\sin 2x + \sin 3x = \cos x - 3\cos 2x + \cos 3x$ is

(JEE – 1989, 2M)

(A) $n\pi + \frac{\pi}{8}$
 (C) $(-1)^n \frac{n\pi}{2} + \frac{\pi}{8}$

(B) $\frac{n\pi}{2} + \frac{\pi}{8}$
 (D) $2n\pi + \cos^{-1} \frac{3}{2}$

Ans: C

The possible values of $\theta \in (0, \pi)$ such that $\sin(\theta) + \sin(4\theta) + \sin(7\theta) = 0$ are

[AIEEE-2011]

(A) $\frac{2\pi}{9}, \frac{\pi}{4}, \frac{\pi}{2}, \frac{2\pi}{3}, \frac{3\pi}{4}, \frac{8\pi}{9}$

(C) $\frac{\pi}{4}, \frac{5\pi}{12}, \frac{\pi}{2}, \frac{2\pi}{3}, \frac{3\pi}{4}, \frac{8\pi}{9}$

(B) $\frac{2\pi}{9}, \frac{\pi}{4}, \frac{4\pi}{9}, \frac{\pi}{2}, \frac{3\pi}{4}, \frac{8\pi}{9}$

(D) $\frac{2\pi}{9}, \frac{\pi}{4}, \frac{\pi}{2}, \frac{2\pi}{3}, \frac{3\pi}{4}, \frac{35\pi}{36}$

Ans: B

DIBY-24

The number of distinct solutions of the equation

$$\frac{5}{4}\cos^2 2x + \cos^4 x + \sin^4 x + \cos^6 x + \sin^6 x = 2 \text{ in the interval } [0, 2\pi] \text{ is}$$

[JEE (Adv.)-2015]

Ans: 08

DIBY-25

The number of solutions of the equation $\sin x = \cos^2 x$ in the interval $(0, 10)$ is

[JEE Mains-2022]

Ans: 04

Consider the following lists :

[JEE (Adv.)-2022]

List-I		List-II	
(I)	$\left\{x \in \left[-\frac{2\pi}{3}, \frac{2\pi}{3}\right] : \cos x + \sin x = 1\right\}$	(P)	has two elements
(II)	$\left\{x \in \left[-\frac{5\pi}{18}, \frac{5\pi}{18}\right] : \sqrt{3}\tan 3x = 1\right\}$	(Q)	has three elements
(III)	$\left\{x \in \left[-\frac{6\pi}{5}, \frac{6\pi}{5}\right] : 2\cos(2x) = \sqrt{3}\right\}$	(R)	has four elements
(IV)	$\left\{x \in \left[-\frac{7\pi}{4}, \frac{7\pi}{4}\right] : \sin x - \cos x = 1\right\}$	(S)	has five elements
		(T)	has six elements

Ans: B

The correct option is:

- (A) (I) $\rightarrow$ (P); (II) $\rightarrow$ (S); (III) $\rightarrow$ (P); (IV) $\rightarrow$ (S)
(B) (I) $\rightarrow$ (P); (II) $\rightarrow$ (P); (III) $\rightarrow$ (T); (IV) $\rightarrow$ (R)
(C) (I) $\rightarrow$ (Q); (II) $\rightarrow$ (P); (III) $\rightarrow$ (T); (IV) $\rightarrow$ (S)
(D) (I) $\rightarrow$ (Q); (II) $\rightarrow$ (S); (III) $\rightarrow$ (P); (IV) $\rightarrow$ (R)

DIBY-27



If the solution of the equation $\log_{\cos x} \cot x + 4\log_{\sin x} \tan x = 1, x \in \left(0, \frac{\pi}{2}\right)$, is

$\sin^{-1}\left(\frac{\alpha + \sqrt{\beta}}{2}\right)$, where α, β are integers, then $\alpha + \beta$ is equal to:

[JEE Mains-2023]

(A) 3

(B) 5

(C) 6

(D) 4

Ans: D

DIBY-28

Let S be the set of all $\alpha \in R$ such that the equation, $\cos 2x + \alpha \sin x = 2\alpha - 7$ has a solution. Then S is equal to:

[JEE Mains-2019]

(A) $[2, 6]$

(B) $[3, 7]$

(C) R

(D) $[1, 4]$

Ans: A

DIBY-29



If α , $-\frac{\pi}{2} < \alpha < \frac{\pi}{2}$ is the solution of $4\cos \theta + 5\sin \theta = 1$, then the value of $\tan \alpha$ is

[JEE Mains-2024]

(A) $\frac{10-\sqrt{10}}{6}$

(B) $\frac{10-\sqrt{10}}{12}$

(C) $\frac{\sqrt{10}-10}{12}$

(D) $\frac{\sqrt{10}-10}{6}$

Ans: C

DIBY-30

The sum of all values of $\theta \in \left(0, \frac{\pi}{2}\right)$ satisfying $\sin^2 2\theta + \cos^4 2\theta = \frac{3}{4}$ is:

[JEE Mains-2019]

(A) π

(B) $\frac{5\pi}{4}$

(C) $\frac{\pi}{2}$

(D) $\frac{3\pi}{8}$

Ans: C

DIBY-31

Let $S = \{\theta \in [-2\pi, 2\pi] : 2\cos^2 \theta + 3\sin \theta = 0\}$. Then the sum of the elements of S is

[JEE Mains-2019]

(A) $\frac{13\pi}{6}$

(B) π

(C) 2π

(D) $\frac{5\pi}{2}$

Ans: C

DIBY-32

Let the set of all $a \in R$ such that the equation $\cos 2x + a\sin x = 2a - 7$ has a solution be

$[p, q]$ and $r = \tan 9^\circ - \tan 27^\circ - \frac{1}{\cot 63^\circ} + \tan 81^\circ$, then pqr is equal to [JEE Mains-2024]

Ans: 48

DIBY-33

Solve: $\sin \theta + \sqrt{3}\cos \theta \geq 1, -\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}$

Ans: $-\frac{\pi}{6} \leq \theta \leq \frac{\pi}{2}$

DIBY-34

Find the value of x in the interval $\left[-\frac{\pi}{2}, \frac{3\pi}{2}\right]$ for which $\sqrt{2}\sin 2x + 1 \leq 2\sin x + \sqrt{2}\cos x$

Ans: $x \in \left[-\frac{\pi}{4}, \frac{\pi}{6}\right] \cup \left[\frac{\pi}{4}, \frac{5\pi}{6}\right]$

DIBY-35

Solve $\sin 2x > \sqrt{2}\sin^2 x + (2 - \sqrt{2})\cos^2 x$

Ans: $\left(n\pi + \frac{\pi}{8}, n\pi + \frac{\pi}{4}\right), n \in \mathbb{I}$

DIBY-36

The set of all x in $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ satisfying $|4\sin x - 1| < \sqrt{5}$ is given by

(A) $\left(-\frac{\pi}{10}, \frac{3\pi}{10}\right)$

(B) $\left(\frac{\pi}{10}, \frac{-3\pi}{10}\right)$

(C) $\left(\frac{\pi}{10}, \frac{3\pi}{10}\right)$

(D) None of these

Ans: A

All the pairs (x, y) that satisfy the inequality $\left(2^{\sqrt{\sin^2 x - 2\sin x + 5}}\right) \cdot \frac{1}{4^{\sin^2 y}} \leq 1$ also satisfy the equation

- A** $2|\sin x| = 2\sin y$
- B** $2\sin x = \sin y$
- C** $\sin x = 2\sin y$
- D** $\sin x = |\sin y|$

Ans: D

DIBY-38

Solve for x & y : $x \cos^3 y + 3x \cos y \sin^2 y = 14$
 $x \sin^3 y + 3x \cos^2 y \sin y = 13$

Ans: $x = \pm 5\sqrt{5}$ & $y = n\pi + \tan^{-1} \frac{1}{2}$

DIBY-39

Solve for x & y : $12\sin x - 2y^2 = 21 - 8y - 5\cos x$

Ans: $x = 2n\pi + \cos^{-1} \frac{5}{13}$ & $y = 2$

DIBY-40

Solve for x & y : $x^2 + 2x\sin(xy) + 1 = 0$

Ans: $x = \pm 1, y = (4n + 3)\frac{\pi}{2}$

DIBY-41



$$\text{Solve } \left(\sin^2 x + \frac{1}{\sin^2 x} \right)^2 + \left(\cos^2 x + \frac{1}{\cos^2 x} \right)^2 = 12 + \frac{1}{2} \sin y$$

$$\text{Ans: } x = k\pi \pm \frac{\pi}{4}; y = 2n\pi + \frac{\pi}{2}; k, n \in I$$

DIBY-42

$$\text{Solve } x \text{ and } y : 4^{\sin x} + 3^{\frac{1}{\cos y}} = 11$$

$$5 \cdot 16^{\sin x} - 2 \cdot 3^{\frac{1}{\cos y}} = 2$$

$$\text{Ans: } x = n\pi + (-1)^n \frac{\pi}{6} \text{ and } y = 2m\pi \pm \frac{\pi}{3}, \text{ where } m, n \in I$$

*It is never about **end result**
It's always about the **Journey***

THANKYOU

#future **IITians**